

Understanding Digital data and storage systems

Digital data refers to information represented in a binary format (ones and zeros) that computers can understand and process. This includes everything from text documents and images to audio and video files, as well as the underlying data for applications and operating systems. Data storage is the technology and methods used to retain digital information for future use. It involves various storage devices and media, such as hard drives, SSDs, cloud storage, and more. Data storage is distinct from computer memory (RAM), which is used for temporary storage and processing while the computer is on.

Digital data storage is the process of saving and preserving digital information using various technologies and media. It encompasses the hardware and software components that enable computers and other devices to store, manage, and retrieve data. Understanding digital data storage is crucial for businesses and individuals alike, as it underpins how we store and access everything from personal photos to large datasets.

File Storage: Data is organized into files, typically with metadata (information about the file).

Block Storage: Data is stored in fixed-size blocks, often used for databases and virtual machines.

Object Storage: Data is stored as objects, with associated metadata, suitable for large amounts of unstructured data.

Data Integrity: Ensuring data is accurate and reliable is a core concern in data storage.

Data Security: Protecting data from unauthorized access and cyber threats is paramount.

Data Accessibility: Making data easily available when needed is crucial for efficient use.

Types of Storage Devices:

Hard Disk Drives (HDDs): Traditional storage devices with spinning disks and read/write heads.

Solid State Drives (SSDs): Faster storage devices using flash memory.

Cloud Storage: Data is stored on remote servers accessed over the internet.

Optical Storage (CDs, DVDs, Blu-ray): Data is stored on optical discs.

Magnetic Tape: Used for archival and backup purposes, often in large data centers.

Understanding boot sequence

The boot sequence, or boot process, is a series of steps a computer takes when powered on or restarted, initializing hardware and loading the operating system. This process ensures

the computer is ready for user interaction by checking vital components, finding the operating system on storage, and loading it into memory. Boot sequence refers to the specific order in which a computer's hardware components are initialized during the boot process.

Steps in Boot Sequence

Power On: The computer receives power, and the BIOS (or UEFI) firmware is activated.

POST (Power-On Self-Test): The BIOS performs a series of tests on the computer's hardware components to ensure they are functioning correctly.

Boot Device Selection: The BIOS searches for a bootable device (e.g., hard drive, SSD, USB drive) based on a pre-defined order.

Boot loader Loading: If a bootable device is found, the BIOS loads the boot loader (a small program that loads the OS) into memory.

Operating System Loading: The boot loader locates and loads the operating system kernel into memory.

Kernel Initialization: The OS kernel initializes and takes control of the system, loading necessary drivers and system processes.

User Login: Finally, the user interface (desktop environment or command line) is loaded, allowing the user to log in and interact with the system.

Understanding Disk Drives

A disk drive is a computer component that reads from and writes data to a disk. The most common type is a hard disk drive (HDD), which uses spinning platters and a read/write head to store and retrieve data. Other types include solid-state drives (SSDs), which use flash memory, and optical drives, which use lasers.

HDDs consist of spinning platters coated with a magnetic material and a read/write head that moves across the platters to access data.

Platters: The platters are rigid and divided into tracks and sectors where data is stored.

Actuator Arm: The actuator arm moves the read/write head to the correct location on the platter to read or write data.

Data Storage: Data is stored as magnetic patterns on the platter's surface.

Interfaces: Common interfaces for HDDs include SATA and SAS.

For fast performance and frequent data access (like gaming, video editing, or the operating system), an SSD is the superior choice. For large storage capacity at a lower cost (like archiving or storing extensive multimedia libraries), an HDD remains a viable and more economical option.

Solid-state Storage Devices (SSDs)

A Solid-State Drive (SSD) is a non-volatile storage device that stores data without using any moving parts, unlike traditional Hard Disk Drives (HDDs), which have spinning disks and mechanical read/write heads. Because of this, SSDs are much faster, more durable, and quieter than HDDs. They load files, programs, and operating systems quickly, making computers more responsive. Since there are no moving parts, SSDs are also less likely to get damaged from drops or shocks. They use less power, which helps extend battery life in laptops. SSDs store data using non-volatile memory, meaning they retain information even when the power is off.

Types of Solid-State Drives

PCIe-based Flash SSDs: These SSDs use the PCI Express (PCIe) connection, which allows for high-speed data transfer. They have very fast read and write speeds, shorter delays (latency), and handle multiple tasks efficiently.

NVMe SSDs: These SSDs use the Non-Volatile Memory Express (NVMe) protocol, designed for super-fast performance. They work with PCIe to achieve speeds over 3000 MB/s, making them perfect for gamers and high-performance tasks. Many come with heatsinks to prevent overheating.

Flash DIMMs: These SSDs are designed as memory modules (DIMMs) instead of traditional storage drives. They are even faster than PCIe SSDs because they avoid PCIe bus conflicts, but they require special drivers and motherboard modifications.

Hybrid DRAM-Flash Storage: These combine flash storage with server DRAM (Dynamic Random Access Memory) for improved performance. This setup helps balance speed and storage capacity efficiently.

Microsoft File Systems

Microsoft's primary file systems are NTFS (New Technology File System) and ReFS (Resilient File System). NTFS is the older, more established system used in most Windows installations, while ReFS is a newer file system designed for increased data integrity and scalability, particularly for large-scale storage scenarios.

FAT32 (File Allocation Table32)

Introduced: 1995 with Windows 95, as an upgrade to previous FAT versions.

Strengths: High compatibility with various operating systems (Windows, Mac, Linux), making it suitable for portable storage like USB drives and SD cards.

Limitations: Max file size 4GB, max partition size 2TB (or 32 GB on Windows by default), lacks security features like permissions and encryption, and prone to fragmentation.

Ideal Use Cases: Removable drives (USB drives, SD cards) where broad compatibility is prioritized over large file sizes or advanced security features.

NTFS (New Technology File System)

Introduced: 1993 with Windows NT, replacing the older FAT systems.

Strengths: Supports large file sizes and volumes (up to 8 Petabytes), offers advanced security features like file permissions and encryption (EFS), incorporates journaling for data integrity and recovery in case of system crashes, and allows file compression and disk quotas.

Limitations: Primarily supported by Windows operating systems (read-only on Mac, limited Linux support), and not ideal for very small volumes (under 400 MB) due to space overhead.

Ideal Use Cases: Internal hard drives and system drives running on Windows operating systems, especially where data security, reliability, and large storage capacities are crucial.

exFAT (Extended File Allocation Table)

Introduced: 2006, specifically for flash memory devices.

Strengths: Overcomes FAT32's 4GB file size limitation, making it suitable for large files on removable media, compatible with Windows and modern macOS versions.

Limitations: Less compatible than FAT32 (requires software for older Linux systems), and lacks advanced security features like encryption.

Ideal Use Cases: Flash drives and other removable media where large files are expected, and a balance between compatibility and capacity is desired.

ReFS (Resilient File System)

Introduced: 2012 with Windows Server 2012.

Strengths: Designed for data integrity and resilience against corruption, boasts features like checksumming, integrity streams, automatic repair, and scalability for massive datasets (exabyte-sized volumes).

Limitations: Primarily intended for server and enterprise environments, not fully compatible with all Windows versions (especially consumer editions), and lacks certain NTFS features like file-level encryption and compression.

Ideal Use Cases: Server environments, Storage Spaces implementations, large-scale virtualization (Hyper-V), and other scenarios where data integrity and availability are paramount, according to Microsoft Learn.

Disk partitions

Disk partitioning is the process of dividing a single physical hard drive into multiple logical drives, or partitions. Each partition acts like a separate drive, allowing for better organization and management of data. These partitions are listed in the partition table and appear as separate drives to the operating system. The process involves creating a partition table on the disk, which stores information about the location and size of each partition. The operating system reads this table to identify and access the different partitions. Each partition can be formatted with a different file system, allowing for different types of data to be stored on the same drive.

Types of Partitions:

Primary Partition: A basic partition that can hold an operating system.

Extended Partition: A special type of partition that can contain multiple logical partitions.

Logical Partition: Partitions within an extended partition.

Understanding FAT

A FAT partition, or File Allocation Table partition, is a type of file system commonly used for storage devices like USB drives and memory cards. It's known for its simplicity and wide compatibility across different operating systems.

A FAT partition is a way of organizing and storing files on a storage device.

It uses a table called the File Allocation Table to keep track of where files are stored on the disk. The FAT contains entries for each cluster (a block of storage) on the disk. Each entry indicates either the next cluster in a file's chain or marks the end of the file or an unused cluster. FAT has different versions, including FAT12, FAT16, FAT32, and exFAT, each with its own characteristics and limitations.

Cross-platform compatibility: When you need a storage device that can be easily used with different operating systems.

Limited resources: When the device has limited resources and needs a simple file system.

Smaller storage devices: FAT is often used for USB drives, memory cards, and other smaller storage devices.

Key components

Boot Sector: Located at the start of the disk, it contains the bootloader code necessary to start the computer and information about the volume's layout and file system structure.

FAT Region: This region stores the primary File Allocation Table, and typically also a duplicate for redundancy purposes in case the primary table gets damaged.

Root Directory Region: Contains information about the files and directories located in the root directory. In FAT12 and FAT16 systems, the root directory has a fixed maximum size and a specified location.

FAT32 stores the root directory within the data region, allowing it to grow dynamically.

Data Region: This is where the actual file and directory data is stored, divided into clusters.

Understanding NTFS, MFT

NTFS (New Technology File System) is a file system developed by Microsoft, and the Master File Table (MFT) is a crucial component within it. The MFT acts as a database that stores metadata about every file and folder on an NTFS volume, including their locations on the disk.

NTFS (New Technology File System):

NTFS is the primary file system used by Windows operating systems.

It provides features like file security, compression, encryption, and journaling (which helps with data recovery).

It uses a more sophisticated structure than older file systems like FAT, making it suitable for larger storage devices and offering better performance and reliability.

MFT (Master File Table):

The MFT is a system file (named \$MFT) within NTFS that serves as a directory or index for all files and directories on the volume.

Each file and folder on an NTFS volume has a corresponding entry in the MFT, known as an MFT entry or MFT record.

For small files, the MFT entry itself may contain the file data, while larger files have pointers to the data stored in other parts of the disk.

These MFT entries contain metadata about the file, such as its name, size, timestamps (creation, modification, last access), security permissions, and pointers to where the actual file data is stored on the disk. The MFT is critical for NTFS functionality; if the MFT is corrupted or damaged, it can lead to data loss or make the file system unusable.

NTFS reserves space for the MFT to minimize fragmentation and ensure efficient access.

The MFT plays a vital role in the efficient operation of the NTFS file system by acting as a fast index to file information. It is also highly valuable in forensic investigations as it can be used to retrieve information about files, even those that have been deleted, as MFT entries are marked as free but the actual data remains until overwritten.

file attributes, file data

NTFS (New Technology File System) organizes all file and directory information as a collection of attributes. Each file or folder on an NTFS volume is viewed as a set of these attributes, which are stored within the Master File Table (MFT).

NTFS File Attributes:

These attributes describe various aspects of a file or directory. Key attributes include:

\$STANDARD_INFORMATION: Contains standard file information like creation, modification, access, and MFT entry change timestamps, as well as file attributes like read-only, hidden, system, and archive.

\$FILE_NAME: Stores the file or directory name, including both long and short (8.3) names. It also maintains its own set of timestamps.

\$DATA: This is the primary attribute containing the actual content of the file.

\$ATTRIBUTE_LIST: Used for files with many attributes that cannot fit within a single MFT record. It lists the locations of these non-resident attributes.

\$SECURITY_DESCRIPTOR: Contains security information, including permissions (ACLs) and ownership.

\$INDEX_ROOT and **\$INDEX_ALLOCATION:** Used for directories to manage file and subdirectory listings.

Alternate Data Streams (ADSs): A unique feature of NTFS, allowing multiple data streams to be associated with a single file. While the primary data stream is typically unnamed (\$DATA:""), ADSs can have specific names (e.g.,

\$DATA:"Second Stream") and store additional, often hidden, data.

File Data (\$DATA Attribute):

The \$DATA attribute specifically refers to the content or raw data of a file.

Resident Data:

For small files, the entire file data can be stored directly within the \$DATA attribute within the MFT record itself. This is known as resident data.

Non-Resident Data:

For larger files, the \$DATA attribute in the MFT record will contain pointers (data runs) to the actual data, which is stored in clusters elsewhere on the disk. This is known as non- resident data.

NTFS compressed files

NTFS (New Technology File System) compression is a built- in feature of Windows operating systems that allows you to reduce the size of files and folders on your hard drive, freeing up valuable disk space. This is particularly useful for large files containing redundant information, like text documents, log files, or backups. The most significant advantage is the reduction in disk space usage, potentially freeing up to 50% or more on highly compressible files. The CPU is faster than the disk I/O, NTFS compression can improve performance by reducing the amount of data that needs to be read from or written to the disk. Reduced Disk I/O: Less data transfer means reduced wear and tear on the disk and increased cache efficiency.

NTFS encrypted file system

The Encrypting File System (EFS) provides the core file encryption technology used to store encrypted files on NTFS volumes. EFS keeps files safe from intruders who might gain unauthorized physical access to sensitive, stored data (for example, by stealing a portable computer or external disk drive). Users work with encrypted files and folders just as they do with any other files and folders. Encryption is transparent to the user who encrypted the file; the system automatically decrypts the file or folder when the user accesses. When the file is saved, encryption is reapplied.

Users who are not authorized to access the encrypted files or folders transparently receive an "Access denied" message if they try to open, copy, move, or rename the encrypted file or folder. The exact message text may vary depending on application which tries to access the file, because it is related not to user rights for file but to ability of EFS to decrypt file using user's private key. It is transparent for user and any applications. There's no risk for user to forget to encrypt file and leave data unprotected. Once file or folder is marked as encrypted, it will be encrypted in background without interaction with user.

Strong key security. In contrast to other solutions when keys are based on user entered pass-phrase, EFS generates keys which are tolerant to dictionary based attacks.

All encrypting/decrypting processes are performed in kernel mode, excluding the risk of leaving key in paging file, from where it could be possibly extracted.

EFS provides data recovery mechanism which is valuable in business environment, giving an organization an opportunity to restore data even if the employee who encrypted it left the company.

Deleting NTFS file system, ReFS

Transitioning from NTFS to ReFS

Steps to Delete/Format an NTFS Drive to ReFS:

Backup Data: Before proceeding, ensure all important data on the NTFS drive is backed up to another location, as formatting will erase all data on the drive.

Open Disk Management:

Right-click the Start button and select "Disk Management."

Identify the NTFS Drive: Locate the drive you wish to format in the Disk Management window.

Format the Drive:

Right-click on the desired NTFS volume. Select "Format..."

Configure Format Options:

In the "File system" dropdown menu, select "ReFS."

Optionally, you can change the "Volume label" and "Allocation unit size."

Ensure "Perform a quick format" is checked for faster formatting.

Confirm and Start:

Click "OK" to confirm the format.

A warning message will appear about data loss; click "OK" to proceed.

The drive will then be formatted with the ReFS file system.

Windows operating systems cannot boot from ReFS volumes; therefore, the primary operating system drive must remain NTFS.

Whole disk encryption

Whole disk encryption, also known as full disk encryption (FDE), is a security measure that encrypts all data on a storage drive, including the operating system, files, and applications. This prevents unauthorized access to the data if the device is lost, stolen, or accessed without proper authentication. FDE ensures that all data on the disk is unreadable without the correct decryption key. FDE uses encryption software or hardware to encrypt every sector of the hard drive or SSD. This process converts the data into an unreadable format, making it inaccessible to anyone without the decryption key.

When the system boots, the user must provide the encryption key (usually a password) to decrypt the drive and access the data.

If the device is compromised, the encrypted data remains protected, as it cannot be accessed without the key.

FDE safeguards sensitive information from unauthorized access, especially in cases of device loss or theft. It helps organizations meet data protection regulations by securing sensitive data. FDE typically operates transparently, meaning users don't have to manually encrypt or decrypt files. By encrypting the entire drive, FDE minimizes the risk of data breaches and unauthorized access to sensitive information. FDE protects data even when the device is turned off.

Example:

Windows BitLocker:

A built-in full disk encryption feature in Windows that allows users to encrypt their entire hard drive.

macOS FileVault:

A built-in full disk encryption tool for macOS that protects data on the hard drive.

Microsoft BitLocker

BitLocker is a Windows security feature that protects your data by encrypting your drives. This encryption ensures that if someone tries to access a disk offline, they won't be able to read any of its content. BitLocker is particularly valuable if your device is lost or stolen, as it keeps your sensitive information secure. It's designed to be user-friendly and integrates seamlessly with the Windows operating system, making it easy to set up and manage.

BitLocker offers two functionalities:

Device Encryption , which is designed for simplicity of use, and it's usually enabled automatically

BitLocker Drive Encryption , which is designed for advanced scenarios, and it allows you to manually encrypt drives

BitLocker provides maximum protection when used with a Trusted Platform Module (TPM), which is a common hardware component installed on Windows devices.

The TPM works with BitLocker to ensure that a device hasn't been tampered with while the system is offline. In addition to the TPM, BitLocker can lock the normal startup process until the user supplies a personal identification number (PIN) or inserts a removable device that contains a startup key. These security measures provide multifactor authentication and assurance that the device can't start or resume from hibernation until the correct PIN or startup key is presented.

On devices that don't have a TPM, BitLocker can still be used to encrypt the operating system drive. This implementation requires the user to either:

Use a startup key, which is a file stored on a removable drive that is used to start the device, or when resuming from hibernation.

Use a password. This option isn't secure since it's subject to brute force attacks as there isn't a password lockout logic. As such, the password option is discouraged and disabled by default.

For BitLocker to use the system integrity check provided by a TPM, the device must have TPM 1.2 or later versions. If a device doesn't have a TPM, saving a startup key on a removable drive is mandatory when enabling BitLocker.

The system BIOS or UEFI firmware (for TPM and non-TPM devices) must support the USB mass storage device class, and reading files on a USB drive in the preboot environment.

Understanding Windows Registry

The Registry or Windows Registry is the database that stores the low-level settings of the operating system and its applications that support registries. It contains all the information of the software and hardware installed on the system. Users can access and configure the settings of Registries by launching the Registry Editor tool of Windows OS. The registries can be considered as the DNA of the Windows devices. It contains commands for the installed applications. Whenever you install a registry-supported Windows application on your system, it automatically adds a new registry entry and stores all the essential information required for running the new application in it

Registries are available to edit on Windows 10/8/7 and up to devices with Windows 95 operating system. To access the registries of different programs installed on your device, Windows provides a graphical interface known as Registry Editor. Registry Editor is not an application that you need to install. It is one of the Windows in-built tools that can be easily accessed through the Run Command Window.

Implementation steps Step-1 :

On your Windows OS system, type Run and click on its icon to launch it.

Step-2 :

The Run Command Box would be launched. Type the Regedit command in the space provided and hit enter to launch the registry editor.

Step-3 :

Look for the registries of the application that you want to change.

Backup of the registries :

To take a backup of the registries, follow these steps as follows.

Open the Registry Editor window

Click on the File menu and select Export.

Choose a safe location on your hard drive or external hard drive and save the registries there.

Registry Root Keys :

When you first launch the Registry Editor, you will notice the Root Keys, containing all different registry values.

HKEY_CLASSES_ROOT (HKCR) It describes the file type, its extension, and Object Linking and Embedding (OLE) information.

HKEY_CURRENT_USER (HKCU) Contains the information and the settings of the currently logged-in user.

HKEY_CURRENT_CONFIG (HKCC) This root key contains the details of the configuration of the hardware currently attached to the device.

Microsoft Windows startup tasks

Windows startup tasks, also known as startup programs, are applications that automatically launch when Windows boots up. Managing these tasks can improve boot times and system performance. Windows provides several ways to view, manage, and configure these tasks.

How to Manage Startup Tasks:

1. Task Manager:

Right-click the taskbar and select Task Manager. Go to the "Startup" tab.

Here, you'll see a list of startup programs, their status (enabled/disabled), and their impact on startup time (None, Low, Medium, High)

You can enable or disable programs by right-clicking and selecting the appropriate option.

2. Settings App:

Open the Settings app (Windows key + I). Navigate to "Apps" then "Startup".

Use the toggles to enable or disable apps. Startup Folder:

Press Windows Key + R, type shell:startup, and press Enter.

This opens the user-specific Startup folder, where you can add shortcuts to programs you want to launch on startup.

System Configuration (msconfig):

Press Windows Key + R, type msconfig, and press Enter. Go to the "Startup" tab to view and manage startup items.

Task Scheduler:

Press Windows Key + R, type taskschd.msc, and press Enter.

This opens Task Scheduler, where you can view, create, and modify tasks including those that run on startup.

Improved Boot Time:

Reducing the number of programs launching at startup can significantly decrease boot time.

Better Performance:

Fewer running programs mean more system resources are available for other tasks, leading to better overall performance.

Control:

You can choose which programs start automatically, preventing unwanted apps from launching when you don't need them.

Encrypting hard drive partitions and USB drives to prevent firm-level attacks: a practical guide

Encrypting data, particularly on devices like hard drive partitions and USB flash drives, is a crucial step in preventing unauthorized access and mitigating the risks associated with data breaches at a firm level.

Why encrypting partitions and drives is important for firm-level security

Protection against theft/loss: Laptops and USB drives are easily lost or stolen, and if they contain sensitive information, this can lead to serious consequences, including financial and legal ramifications for the company. Encrypting the data renders it useless to anyone without the decryption key.

Mitigating insider threats: While companies invest in perimeter security, malicious actors can still compromise data by gaining access through employees. Encrypting data on employee devices adds an extra layer of security, especially when combined with strong access controls and monitoring tools.

Safeguarding intellectual property: Trade secrets, proprietary algorithms, and customer data are critical assets for businesses. Encryption helps protect these assets from unauthorized access and corporate espionage.

Compliance with regulations: Many industries have strict regulations regarding data protection (e.g., HIPAA, PCI DSS, GDPR). Encryption can help companies meet these compliance requirements and avoid potential fines.

Protection against firmware-level attacks: Firmware attacks, where malicious code is injected into a device's low-level software, can bypass traditional security measures. Encrypting the entire drive or using self-encrypting drives with hardware-level encryption can make it harder for attackers to compromise and extract data.

How to encrypt a partition or USB drive:

1. Choose encryption software:

Popular options include BitLocker (for Windows) and FileVault (for macOS).

2. Enable encryption:

Follow the instructions provided by your chosen software. For BitLocker, you'll typically right-click on the drive, select "Turn on BitLocker", and choose a strong password according to Microsoft.

3. For USB drives:

Encrypt the entire drive or a specific partition containing sensitive data.

4. For hard drive partitions:

You can encrypt individual partitions or the entire hard drive, depending on your needs and the software capabilities.

Strong passwords/passphrases:

Use strong, unique passwords or passphrases to protect your encryption keys.

Backup recovery keys:

Ensure you have backups of your recovery keys in case you forget your password or need to access the data in an emergency.

Regular updates:

Keep your encryption software and operating system updated to protect against vulnerabilities.